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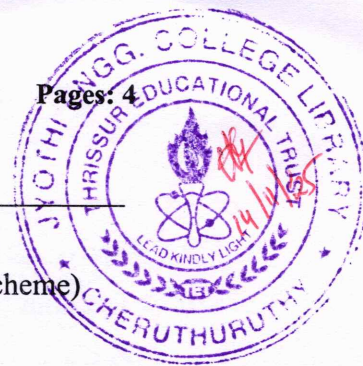
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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R) (FT/WP) Examination November 2025 (2024 Scheme)

**Course Code: GAMAT301****Course Name: MATHEMATICS FOR INFORMATION SCIENCE-3**

Max. Marks: 60

Duration: 2hours 30minutes

PART A*(Answer all questions. Each question carries 3 marks)*

		CO	Marks
1	Find the probability distribution function of a Binomial distribution, given that the number of trials is 5 and the sum of its mean and variance is 1.8.	1	(3)
2	A discrete random variable X has the probability distribution $P(X = x) = \frac{k}{2^x}; x = 0, 1, 2, 3, 4.$ Find (i) the value of k (ii) the probability that X takes an even value.	1	(3)
3	The cumulative distribution function of a continuous random variable X is defined by: $F(x) = \begin{cases} 0 & , x \leq 2 \\ k(x-2) & , 2 < x < 6. \\ 1 & , x \geq 6 \end{cases}$ Find (i) the value of k . (ii) $P(X > 4)$	2	(3)
4	If X follows an exponential distribution with $P(X \leq 1) = P(X > 1)$, find mean and variance of X .	2	(3)
5	Define and provide examples for the classification of random processes.	3	(3)
6	Let $X_1, X_2, \dots, X_{10}$ be independent Poisson random variables with mean 1. Use the Markov inequality to get a bound on $P(X_1 + X_2 + \dots + X_{10} \geq 15)$.	3	(3)
7	State Chapman-Kolmogorov theorem in homogeneous Markov Chain.	4	(3)
8	If the transition probability matrix of a Markov chain is $P = \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix}$, find the steady state distribution of the chain.	4	(3)

PART B*(Answer any one full question from each module, each question carries 9 marks)*

Module -1

- 9 a) Derive the mean and variance of the Poisson Distribution. 1 (4)
- b) The joint probability mass function of two random variables X and Y is given by 1 (5)
- $$P(x, y) = k(x + 2y); x = 0, 1, 2; y = 0, 1, 2, 3.$$
- Find (i) the value of k.
 (ii) marginal density functions of X and Y.
 (iii) $P(X+Y \leq 3)$
- 10 a) In an examination, a candidate is required to answer 15 multiple-choice questions, 1 (5)
- each having four possible options. He knows the correct answers to 10 of these questions, and for the remaining 5 questions, he selects an answer at random.
- (i) What is the probability that he answers 13 or more questions correctly?
 (ii) What is the mean and variance of the total number of correct answers he gives?
- b) It is known that 2% of the bolts produced by a company are defective. The bolts 1 (4)
- are supplied in boxes of 200 bolts. What is the probability that a randomly chosen box contains no more than 5 defective bolts? In a consignment of 1000 such boxes, how many can be expected to have more than 5 defective bolts?

Module -2

- 11 a) X is uniformly distributed with mean 1 and variance $\frac{4}{3}$. If three independent 2 (5)
- observations of X are made, what is the probability that all of them are negative?
- b) The life times of tube light bulbs produced by a company are normally distributed 2 (4)
- with mean 1000 hrs and standard deviation 100 hrs. Is this company correct when it claims that 95% of its light bulbs last at least 900 hrs?
- 12 a) The joint probability distribution function of two continuous random variable X 2 (5)
- and Y is:
- $$f(x, y) = \begin{cases} 8xy, & 0 < y < x < 1 \\ 0, & \text{otherwise} \end{cases}$$
- (i) Find $P(X+Y < 1)$ (ii) Check whether X and Y are independent
- b) Suppose a new machine is put into operation at time zero. Its life time is an 2 (4)
- exponential random variable with mean life 12 hrs.
- (i) What is the probability that the machine will work continuously for one day?

- (ii) Suppose the machine has not failed by the end of the first day, what is the probability that it will work for the whole of the next day?

Module -3

- 13 a) In a game involving repeated throws of a fair die, a person receives ₹3 if the number obtained is greater than or equal to 3, and loses ₹3 otherwise. Using the Central Limit Theorem, find the probability that after 25 throws, the person's total earnings exceed ₹25. 3 (5)
- b) A radioactive source emits particles at a rate of 5 per minute in accordance with a Poisson process. Each particle emitted has a probability of 0.6 of being recorded by a device. 3 (4)
- (i) Find the probability that 10 particles are recorded in a four-minute period.
- (ii) On the average how many particles go unrecorded in a one-hour period?
- (iii) What is the expected time until 10th the particle is recorded?
- 14 a) Suppose the number of items produced in a factory during a week is a random variable with a mean of 500 and a variance of 100. What is the probability that the production in a given week will be between 400 and 600? 3 (4)
- b) The number of failures occurring in a computer network follows a Poisson process. On average, one failure occurs every four hours. Find the probability of: 3 (5)
- (i) At most one failure occurring in the first 8 hours.
- (ii) At most one failure occurring in the first 8 hours and at least two failures occurring in the next 8 hours.

Module -4

- 15 a) Three boys — A, B, and C — are throwing a ball to one another. A always throws the ball to B, and B always throws the ball to C, but C is equally likely to throw the ball to either A or B. 4 (5)
- (i) Considering this process as a Markov chain, construct the transition probability matrix.
- (ii) If the ball is initially with C, find the probability that it is with B after two passes.
- (iii) Determine the steady-state (long-run) probabilities of finding the ball with each of them.

- b) Consider a Markov chain with state space $S = \{1, 2, 3, 4, 5\}$ and the following transition matrix: 4 (4)

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ 0.2 & 0.3 & 0.5 & 0 & 0 \\ 0 & 0 & 1.0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1.0 \\ 0 & 0 & 0 & 1.0 & 0 \end{bmatrix}$$

- (i) Identify the communicating classes of the Markov chain.
- (ii) Classify each communicating class as **recurrent** or **transient**.
- (iii) Determine whether the Markov chain is **irreducible** and **aperiodic**.

- 16 a) A Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ is defined on the state space $S = \{1, 2, 3\}$ with the initial distribution 4 (4)

$$P(X_0 = i) = \frac{1}{3}, i = 1, 2, 3; \text{ and TPM } P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}.$$

Find (i) $P(X_2 = 3)$ (ii) $P(X_1 = 1, X_2 = 2, X_3 = 3)$ (iii) $P(X_2 = 2, X_3 = 3 | X_1 = 1)$

- b) A gambler starts with a capital of ₹2 and plays a fair coin-toss betting game. 4 (5)
In each round, he wins ₹1 if a head appears and loses ₹1 if a tail appears.
The gambler stops playing as soon as his capital either increases to ₹4 or reduces to ₹0. Model this situation as a Markov chain and determine the transition probability matrix.

Find

- (i) The probability that the gambler loses all his money at the end of two plays.
- (ii) The probability that game ends exactly after two rounds.
- (iii) The probability that the gambler proceeds to the third round.

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